



Chemistry of the atmosphere

Name: _____

Class: _____

Date: _____

Time:

Marks: **67 marks**

Comments:

1.

Titan is a moon of the planet Saturn.

The table below shows the percentages of some gases in the atmosphere of Titan and in the atmosphere of the Earth.

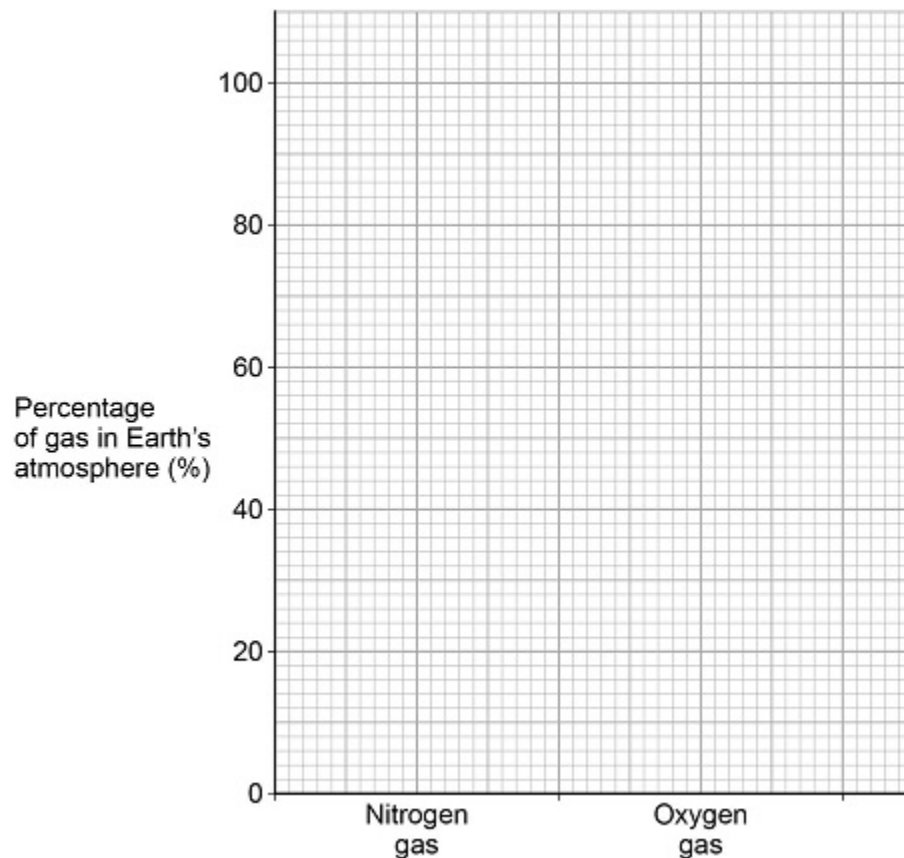
Gas	Percentage of gas in atmosphere (%)	
	Titan	Earth
Nitrogen	98	78
Oxygen	Zero	21
Methane	1.4	0.0002
Argon	0.14	0.9
Carbon dioxide	0.0001	0.04

- (a) Which **two** gases are present in smaller percentages on the Earth than on Titan?

_____ and _____

(1)

- (b) Complete the bar chart in the figure below to show the percentages of nitrogen gas and oxygen gas in the Earth's atmosphere.



(2)

(c) Why are algae less likely to photosynthesise on Titan than Earth?

Use the table above.

Tick (✓) **one** box.

Titan's atmosphere contains too little argon.

☐

Titan's atmosphere contains too little carbon dioxide.

☐

Titan's atmosphere contains too little methane.

☐

Titan's atmosphere contains too little nitrogen.

☐

(1)

(d) Titan is warmer than the other moons of Saturn because of the greenhouse effect.

How do greenhouse gases trap energy from the sun?

Tick (✓) **one** box.

All wavelengths of radiation are reflected back to the surface of Titan.

☐

Long wavelength radiation is reflected back to the surface of Titan.

☐

Short wavelength radiation is reflected back to the surface of Titan.

☐

(1)

As well as methane, the atmosphere of Titan contains small amounts of propene gas. Methane is an alkane and propene is an alkene.

- (e) Bromine water is an orange solution used to identify alkenes.

Draw **one** line from each gas to its effect on bromine water.

Gas	Effect on bromine water
	Forms a blue solution
Methane	Forms a colourless solution
	Forms a green solution
Propene	Forms a white precipitate
	No effect

(2)

- (f) Propene reacts with water (steam) to make propanol.

The ratio of the masses of propene and water that react is:

Propene : water

7 : 3

Calculate the mass of propene that reacts with 21 g water.

Mass = _____ g

(2)

(Total 9 marks)

2.

Scientists study the atmosphere on planets and moons in the Solar System to understand how the Earth's atmosphere has changed.

- (a) Millions of years ago the Earth's atmosphere was probably just like that of Mars today.

The table shows data about the atmosphere of Mars and Earth today.

Mars today		Earth today	
nitrogen	3%	nitrogen	78%
oxygen	trace	oxygen	21%
water	trace	water	trace
Carbon dioxide	95%	Carbon dioxide	trace
Average surface temperature -23°C		Average surface temperature 15°C	

The percentages of some gases in the Earth's atmosphere of millions of years ago have changed to the percentages in the Earth's atmosphere today.

For **two** of these gases describe how the percentages have changed **and** suggest what caused this change.

(2)

- (b) Titan is the largest moon of the planet Saturn.
Titan has an atmosphere that contains mainly nitrogen.
Methane is the other main gas.

Main gases in Titan's atmosphere	Percentage (%)	Boiling point in °C
Nitrogen	95	-196
Methane	5	-164
Average surface temperature -178°C		

When it rains on Titan, it rains methane!

Use the information above and your knowledge and understanding to explain why.

(2)

- (c) Ultraviolet radiation from the Sun produces simple alkenes, such as ethene (C₂H₄) and propene (C₃H₆) from methane in Titan's atmosphere.

State the general formula for alkenes.

(1)

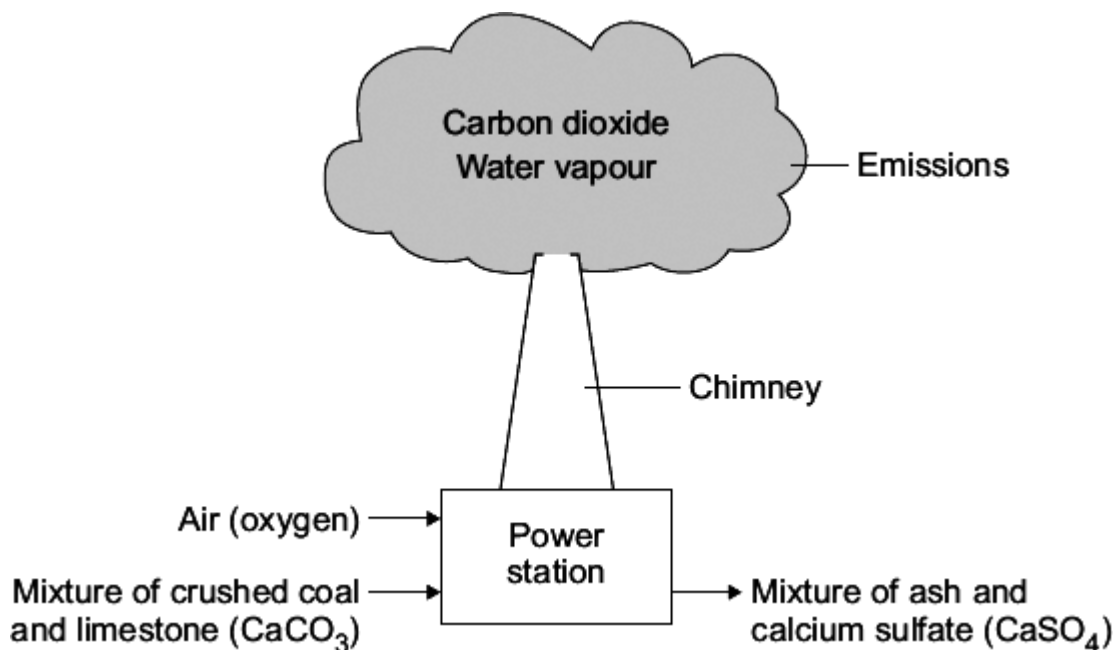
(Total 5 marks)

3.

Most power stations burn coal to generate electricity. Burning coal gives off sulfur dioxide gas which can be removed from the waste gases by using limestone.

This prevents sulfur dioxide from entering the atmosphere and causing acid rain.

One disadvantage of using limestone in a power station is that it releases 'locked up carbon dioxide' into the atmosphere.



(a) How does the limestone used in a power station:

(i) release carbon dioxide

(1)

(ii) remove sulfur dioxide?

(1)

(b) The waste gases from the chimney are monitored. One toxic gas that should not be released is carbon monoxide.

Explain how carbon monoxide would be formed.

(2)

(c) The use of limestone in a power station releases 'locked up carbon dioxide' into the atmosphere.

(i) Explain the meaning of 'locked up carbon dioxide'.

(2)

(ii) Why does the release of this carbon dioxide cause an environmental problem?

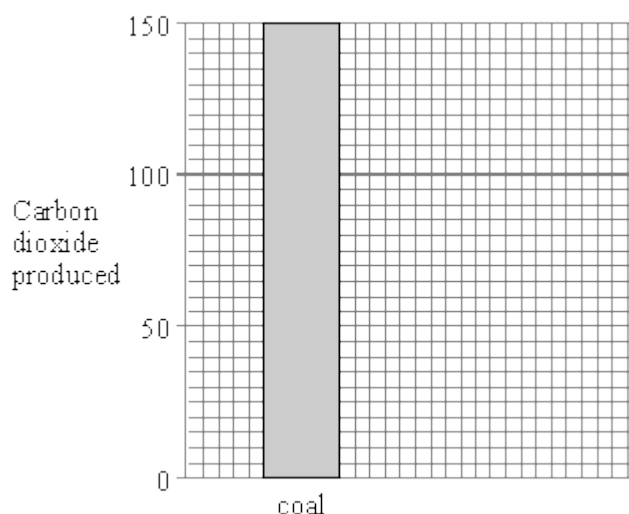
(1)

(Total 7 marks)

4.

The table shows how much carbon dioxide is produced when you transfer the same amount of energy by burning coal, gas and oil.

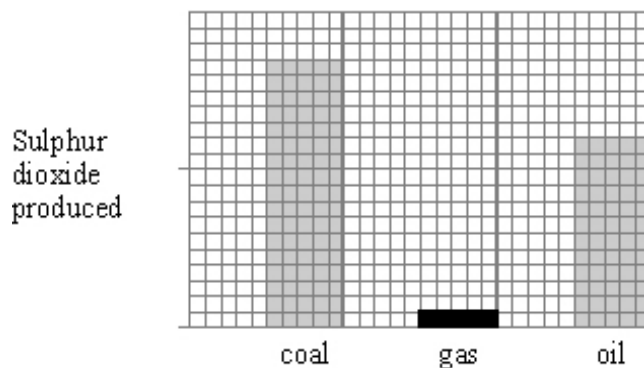
	Carbon dioxide (based on oil = 100)
coal	150
gas	75
oil	100



(a) Use the information from the table to complete the bar-chart.

(2)

- (b) The second bar-chart shows how much sulphur dioxide is produced by burning the same three fuels.



Compare the amount of sulphur produced by burning gas with the amount produced by burning coal.

(1)

- (c) Burning fuels also produces nitrogen oxides, even though the fuels contain no nitrogen. Explain why this happens.

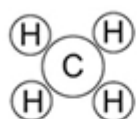
(2)

- (d) When you release the same amount of energy from coal, gas and oil, different amounts of carbon dioxide are produced. Use the information below to explain why.

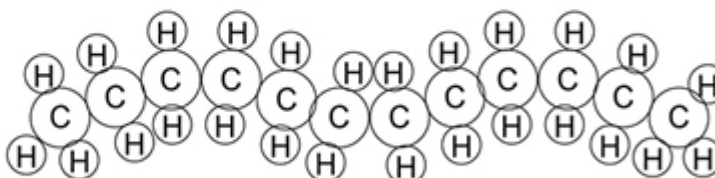
Coal is mainly carbon



North Sea gas is mainly methane



Oil is made from molecules similar to the one shown



(3)

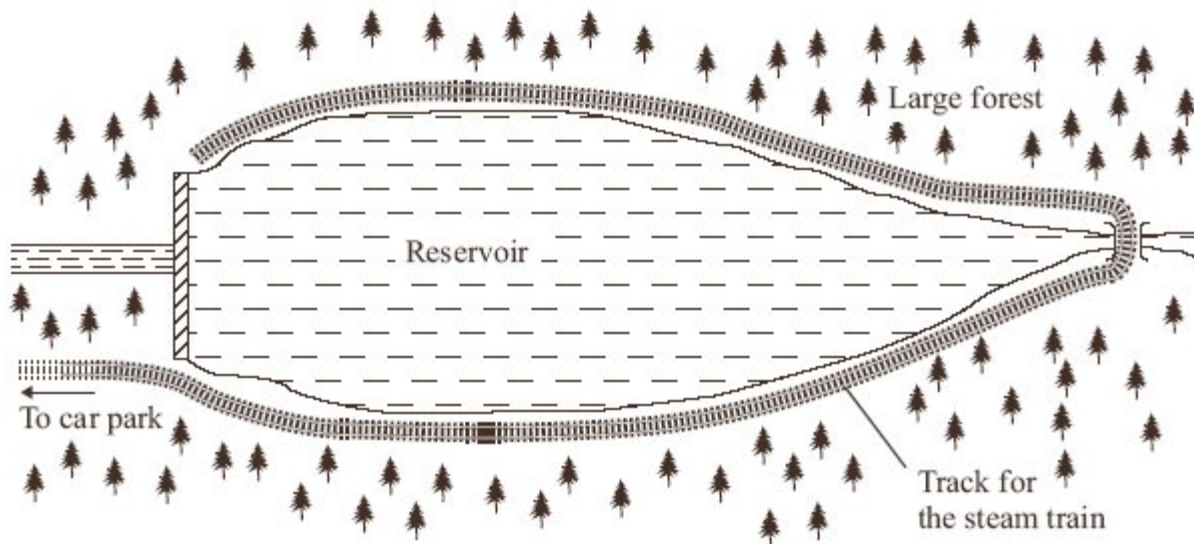
(e) What other element do coal and oil usually contain?

(1)

(Total 9 marks)

5.

A large reservoir is surrounded by trees. Planners need to protect the environment. The distance around the reservoir is many kilometres. There will be only one road access to a car park a few kilometres from the reservoir. From the car park people would be transported to accommodation, activities or places of interest by steam train.



- (a) Coal contains carbon and small amounts of sulfur. The steam train would cause environmental problems if coal were used as the fuel.

Explain why.

(4)

- (b) The planners have stated that, as a result of using the steam train, there must be no overall increase of carbon dioxide added to the atmosphere. The steam train would be considered as 'carbon neutral' if wood, from the surrounding forest, were used as the fuel.

Suggest why.

(3)

(Total 7 marks)

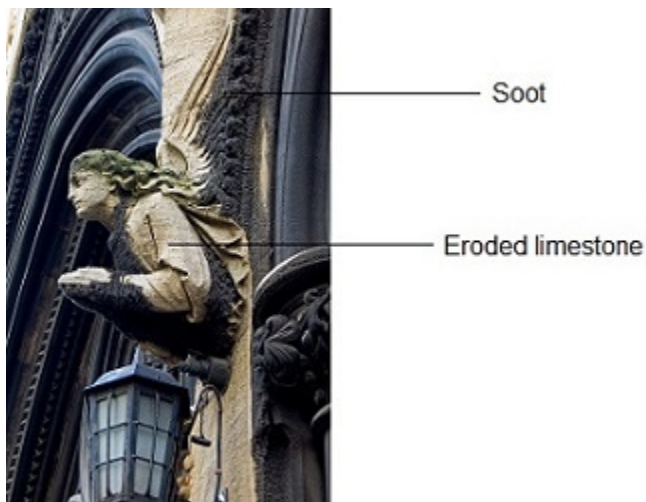
6.

This question is about atmospheric pollution.

The image below shows a limestone carving which has been damaged by atmospheric pollution.

The carving has been:

- blackened by soot
- eroded where the limestone has reacted with atmospheric pollutants.



- (a) Explain why soot is formed when some fossil fuels are burned.

(2)

- (b) Fossil fuels are burned in car engines.

Explain how reducing the amount of sulfur in fossil fuels reduces the erosion of limestone.

(4)

- (c) Oxides of nitrogen are atmospheric pollutants which are formed in car engines.

Explain why oxides of nitrogen are formed in car engines.

(2)

(Total 8 marks)

7.

Sulfur is a non-metal.

Sulfur burns in the air to produce sulfur dioxide, SO_2

- (a) Why is it important that sulfur dioxide is **not** released into the atmosphere?

Tick (✓) **one** box.

Sulfur dioxide causes acid rain.

☐

Sulfur dioxide causes global dimming.

☐

Sulfur dioxide causes global warming.

☐

(1)

- (b) Sulfur dioxide dissolves in water.

What colour is universal indicator in a solution of sulfur dioxide?
Give a reason for your answer.

(2)

- (c) Sulfur dioxide is a gas at room temperature.

The bonding in sulfur dioxide is covalent.

Explain, in terms of its structure and bonding, why sulfur dioxide has a low boiling point.

(3)

- (d) *In this question you will be assessed on using good English, organising information clearly and using specialist terms where appropriate.*

Sulfur dioxide is produced when fossil fuels are burned.

It is important that sulfur dioxide is not released into the atmosphere.

Three of the methods used to remove sulfur dioxide from gases produced when fossil fuels are burned are:

- wet gas desulfurisation (**W**)
- dry gas desulfurisation (**D**)
- seawater gas desulfurisation (**S**).

Information about the three methods is given in the bar chart and in **Table 1** and **Table 2**.

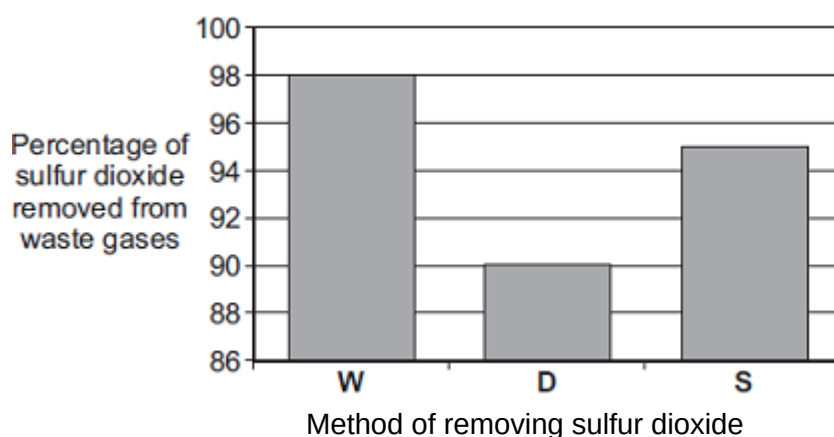


Table 1

Method	Material used	How material is obtained
W	Calcium carbonate, CaCO ₃	Quarrying
D	Calcium oxide, CaO	Thermal decomposition of calcium carbonate: $\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2$
S	Seawater	From the sea

Table 2

Method	What is done with waste material
W	Solid waste is sold for use in buildings. Carbon dioxide is released into the atmosphere.
D	Solid waste is sent to landfill.
S	Liquid waste is returned to the sea.

Evaluate the three methods of removing sulfur dioxide from waste gases.

Compare the three methods and give a justified conclusion.

[illegible]

(6)

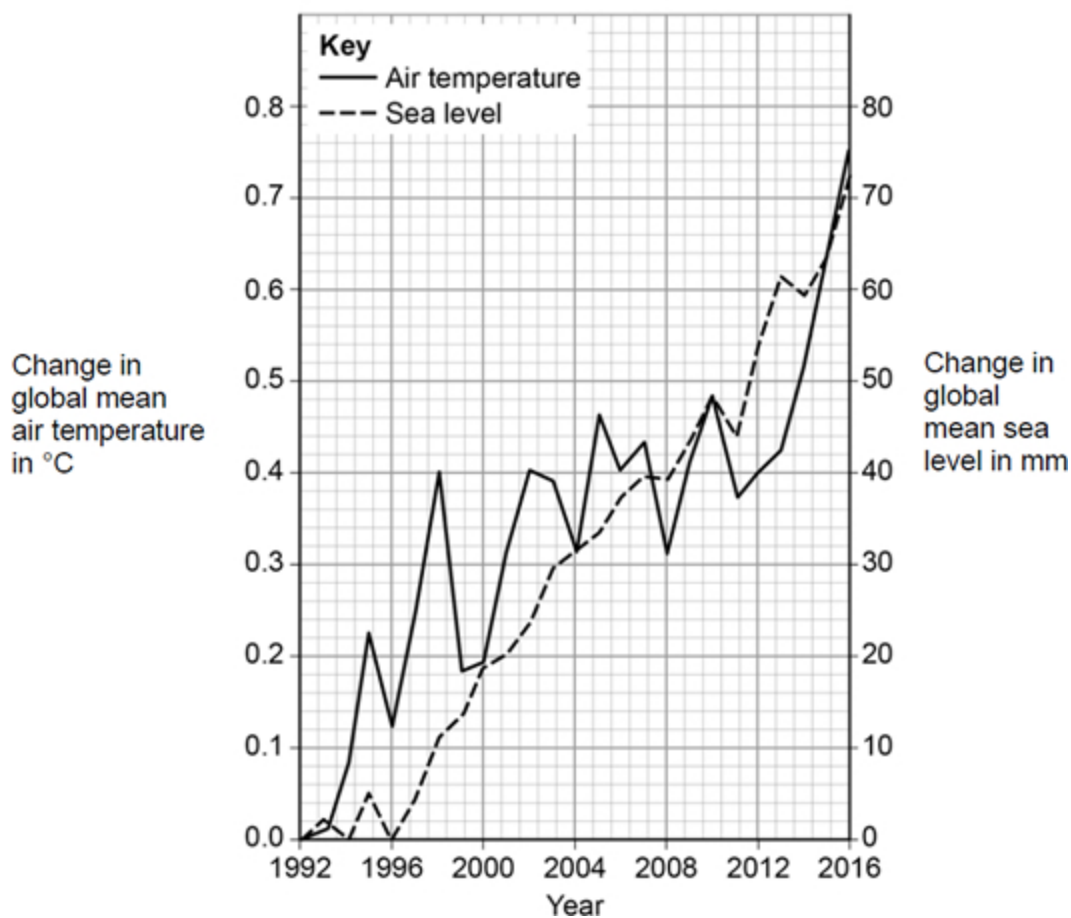
(Total 12 marks)

8.

This question is about climate change.

Figure 1 shows the changes in the global mean air temperature and global mean sea level from 1992 to 2016.

Figure 1



- (a) Calculate the mean yearly increase in sea level between 1992 and 2016.

Use **Figure 1**.

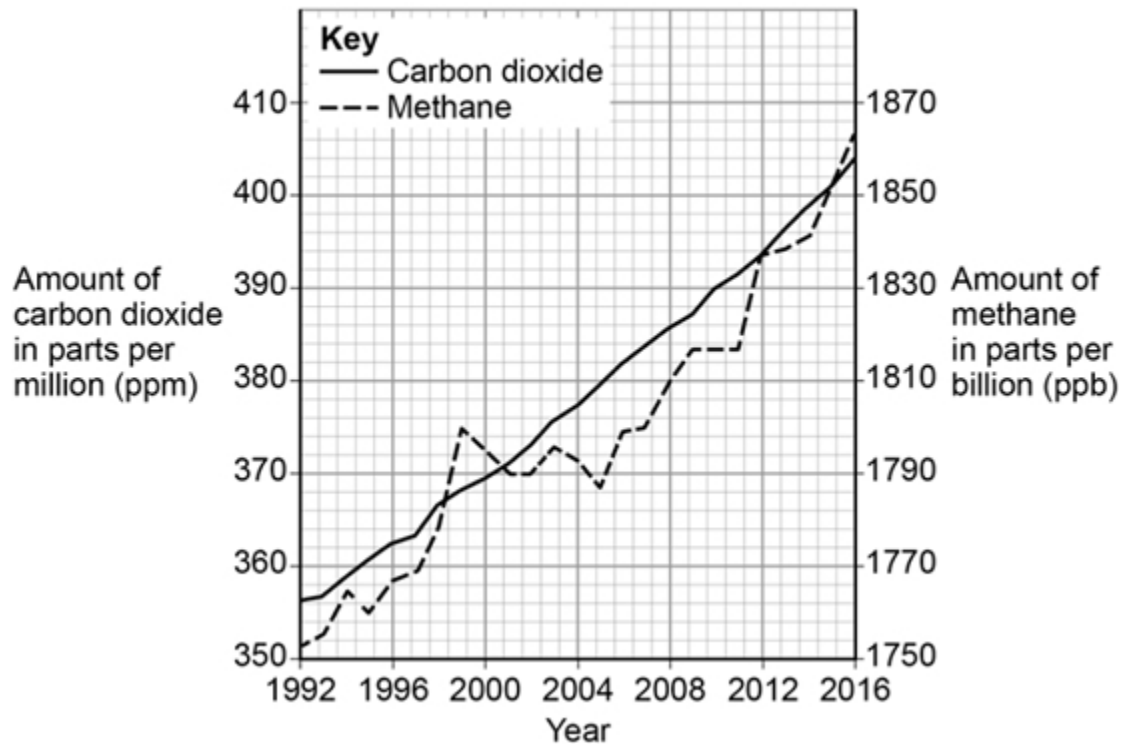
Mean yearly increase in sea level = _____ mm / year

(2)

Most scientists think carbon dioxide and methane are a cause of global climate change.

Figure 2 shows the amounts of these gases in the atmosphere from 1992 to 2016.

Figure 2



(b) Describe the changes in **Figure 1** and in **Figure 2**.

Explain how these changes have taken place.

(6)

(c) The data was collected by a single scientific group.

Give **two** reasons why more evidence is needed to support any conclusions made by this scientific group.

1. _____

2. _____

(2)

(Total 10 marks)

Mark schemes

1.

(a)

in either order, both required for mark
allow phonetic spellings

nitrogen

allow N₂ for nitrogen

and

methane

allow CH₄ for methane

1

(b)

ignore width of bars
ignore additional bars

nitrogen bar to 78%

1

oxygen bar to 21%

1

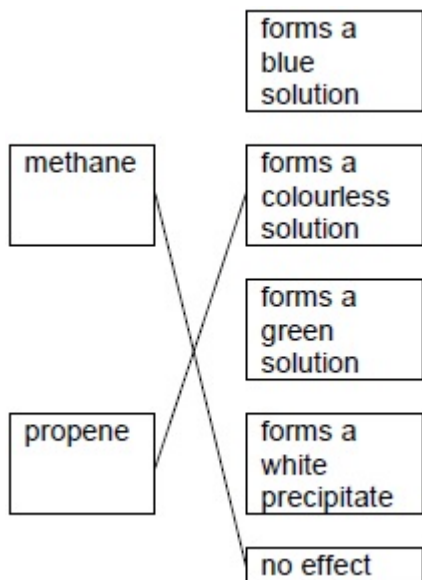
(c) Titan's atmosphere contains too little carbon dioxide.

1

(d) long wavelength radiation is reflected back to the surface of Titan.

1

(e)



an extra line from a gas to an effect on bromine water
negates that mark

2

(f)

an answer of 49 (g) scores 2 marks.

$$\frac{7 \times 21}{3}$$

1

$$= 49 \text{ (g)}$$

1

[9]

2.

(a) any **two** from:

*asks for cause therefore no marks for just describing the change
must link reason to a correct change in a gas*

carbon dioxide has decreased due to:

accept idea of 'used' to indicate a decrease

- plants / microorganisms / bacteria / vegetation / trees
- photosynthesis
ignore respiration
- 'locked up' in (sedimentary) rocks / carbonates / fossil fuels
- dissolved in oceans
ignore volcanoes

oxygen has increased due to:

accept idea of 'given out / produced'

- plants / bacteria / microorganisms / vegetation / trees
- photosynthesis
ignore respiration

nitrogen increased due to:

accept idea of 'given out / produced'

- ammonia reacted with oxygen
- bacteria / micro organisms
ignore (increase in) use of fossil fuels / deforestation

2

(b) (because methane's) boiling point is greater than the average / surface temperature
or Titan's (average / surface) temperature is below methane's boiling point

*ignore references to nitrogen **or** water*

1

any methane that evaporates will condense

accept boils for evaporates

accept cooling and produce rain for condensing

1

(c) C_nH_{2n}

1

[5]

3.

(a) (i) (thermal) decomposition

allow it breaks down

accept symbol equation or in words

allow reaction with SO_2 (to form CO_2)

1

(ii) calcium carbonate / calcium oxide / limestone / quicklime / it reacts with sulfur dioxide / forms calcium sulfate

accept it neutralises sulfur dioxide / neutralisation

ignore references to sulfur

do not accept 'calcium reacts with...'

1

(b) by incomplete / partial combustion (of the fuel)

1

insufficient oxygen / air (to burn fuel)

accept insufficient oxygen / air to burn fuel completely for 2 marks

if no other marks awarded

accept $C + CO_2 \rightarrow 2CO$ or

$2C + O_2 \rightarrow 2CO$ or in words for 1 mark

1

(c) (i) any **two** from:

- (CO_2) from the atmosphere
- (CO_2) taken in millions of years ago **or** early (atmosphere)
allow thousands / billions
allow rocks formed millions of years ago
- (CO_2) was used to form the shells / skeletons of marine organisms / fossil fuels
accept sedimentary rocks
allow used to form correct named fossil fuel
ignore limestone

2

(ii) any **one** from:

- (increases / enhances) global warming
allow greenhouse gas / effect
*do **not** accept ozone layer / acid rain / global dimming*
ignore consequences of global warming
- is additional carbon dioxide **or** not able to be absorbed by oceans / seas **or** used by (green) plants
- acidification of sea water

1

[7]

4.

(a) both bars correct height (to better than half a square)
1 mark for both

both bars correctly labelled
(w.r.t. relative heights if these incorrect)
for 1 mark

2

(b) a lot less / much less / 18 times less (converse must specify coal)
gains 1 mark

1

(c) *ideas that*

- at high temperatures (produced when fuels burn)
- nitrogen and oxygen from air / atmosphere combine / react
or nitrogen from air / atmosphere oxidises
for 1 mark each

2

(d) *ideas that*

- coal produces most carbon dioxide / more CO₂ than gas / oil
- because coal is (mostly) carbon
- gas produces less carbon dioxide than coal / oil
- oil and gas also contain hydrogen / contain more hydrogen atoms than carbon atoms
/ also produce water
any three for 1 mark each

3

(e) sulphur

for 1 mark

2

[9]

5.

(a) any **two** environmental problems with linked explanations

- global warming (1)
accept effects of global warming
caused by (formation of) carbon dioxide / greenhouse gas (1)
ignore greenhouse effect
- acid rain (1)
accept effects of acid rain
ignore respiratory problems
caused by (formation of) sulfur dioxide (1)
accept sulfur oxide
ignore sulfuric acid
- global dimming (1)
ignore respiratory problems
caused by (formation of) particles / particulates / fires /
smoke / carbon / pm 10 (1)
- scarring of landscape (1)
caused by mining / quarrying of coal (1)
ignore ozone layer

max 4

(b) any **three** from:

- replant the trees / renewable / sustainable
ignore reusable
- carbon dioxide is used by the trees / photosynthesis
accept trees absorb carbon dioxide as they grow
*do **not** allow respiration*
- it's a (continuous carbon) cycle
accept 'carbon dioxide goes back into the air'
accept trees use CO₂ which is released when trees are burnt
- no 'new' carbon (dioxide) is produced **or**
no locked up carbon (dioxide) is released
accept no carbon (dioxide) from fossil fuels is produced

3

[7]

6.	(a) incomplete combustion <i>max 1 mark if soot wrongly identified</i>	1
	(because of) insufficient oxygen	1
	(b) sulfur reacts with oxygen to form sulfur dioxide <i>allow SO₂ for sulfur dioxide</i> <i>allow sulfur burns to form sulfur dioxide</i>	1
	(so) less sulfur dioxide emitted	1
	(so) less acid rain	1
	(so less) limestone reacts with acid rain	1
	(c) (car engines work at) high temperatures	1
	(so in the engine) nitrogen (from air) reacts with oxygen (from air)	1
	[8]	
7.	(a) Sulfur dioxide causes acid rain.	1
	(b) red / orange / yellow <i>do not accept any other colours</i>	1
	because sulfur dioxide (when in solution) is an acid	1
	(c) (there are) <u>weak</u> forces (of attraction) <i>do not accept any reference to covalent bonds breaking</i>	1
	between the molecules <i>do not accept any other particles</i>	1
	(these) take little energy to overcome <i>award third mark only if first mark given</i>	1

- (d) Marks awarded for this answer will be determined by the Quality of Communication (QC) as well as the standard of the scientific response. Examiners should also refer to the information on page 5 and apply a 'best-fit' approach to the marking.

0 marks

No relevant content

Level 1 (1 – 2 marks)

A relevant comment is made about the data.

Level 2 (3 – 4 marks)

Relevant comparisons have been made, and an attempt made at a conclusion.

Level 3 (5 – 6 marks)

Relevant, detailed comparisons made and a justified conclusion given.

examples of the points made in the response

effectiveness

- W removes the most sulfur dioxide
- D removes the least sulfur dioxide

material used

- Both W and D use calcium carbonate
- Calcium carbonate is obtained by quarrying which will create scars on landscape / destroy habitats
- D requires thermal decomposition, this requires energy
- D produces carbon dioxide which may cause global warming / climate change
- S uses sea water, this is readily available / cheap

waste materials

- W product can be sold / is useful
- W makes carbon dioxide which may cause global warming / climate change
- D waste fill landfill sites
- S returned to sea / may pollute sea / easy to dispose of

6

[12]

8.

- (a) 72/24

an answer of 3 (mm / year) scores 2 marks

1

= 3 (mm / year)

an answer of 3.125 (mm / year) scores 1 mark

1

(b) **Level 3 (5-6 marks):**

Relevant points (reasons / causes) are identified, given in detail and logically linked to form a clear account.

Level 2 (3-4 marks):

Relevant points (reasons / causes) are identified, and there are attempts at logical linking. The resulting account is not fully clear.

Level 1 (1-2 marks):

Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.

Level 0

No relevant content

Indicative content

description

- global air temperature has risen overall / erratically
- mean sea level has risen (steadily)
- carbon dioxide has risen steadily
- methane has risen overall / erratically

explanations

- (carbon dioxide increase because) increase in fossil fuel combustion
or
- (carbon dioxide increase because) increase in deforestation
- methane from cattle / landfill / rice plantations
- carbon dioxide and / or methane trap heat
or
- carbon dioxide and / or methane are greenhouse gases
- polar ice caps melt
or
- seawater expands

linked explanation

- greenhouse gases linked to temperature rise
- temperature rise linked to seawater level

6

(c) any **two** from:

- bias
- simplified models
- lack of peer review
ignore reproducible

2

[10]

Examiner reports

1.

- (a) Approximately 70% of students correctly interpreted the percentages in the table to identify the gases present in smaller proportions on Earth.
- (b) Generally, drawing the two bars to show the percentage of nitrogen and oxygen in Earth's atmosphere was well done. Some students did, however, use the wrong data.
- (c) Most students realised that it was a lack of carbon dioxide that would prevent photosynthesis on Titan.
- (d) Only around 40% of students chose the correct cause of global warming from the three given.
- (e) This proved to be a difficult question, with only about 15% correctly linking both boxes. Over half scored zero marks.
- (f) Around 70% of students used the proportions correctly to calculate the mass of propene.

2.

- (a) The question of how the percentages of two gases in the Earth's early atmosphere have changed to the percentages in the Earth's atmosphere today was a good discriminator. The two most common correct answers were carbon dioxide and oxygen, both linked to photosynthesis.
- (b) Given the boiling point of methane and the surface temperature on Titan most students could not explain why on Titan, it rains methane. The negative temperatures obviously caused problems for the majority of students.
- (c) The general formula for alkenes was not well known.

3.

- (a)
 - (i) Most candidates found it difficult to express how limestone releases carbon dioxide. Answers such as decomposition were rare and yet burning was more common.
 - (ii) Most candidates found it difficult to express how limestone removes sulfur dioxide. The correct idea of a chemical reaction or neutralisation was not often seen.
- (b) This part was not answered well because many candidates thought that the carbon monoxide came directly from heating or burning limestone.
- (c)
 - (i) This part was poorly answered with very few candidates gaining two marks. The idea of locked up carbon dioxide was not at all well understood. Many candidates thought that the carbon dioxide was locked up in green plants or even locked up in the chimney.
 - (ii) The majority of candidates gained the mark because they knew that carbon dioxide causes global warming.

5.

The stem of part (a) did state that coal contains both carbon and sulfur. However, a small number of candidates only addressed one of them. Many candidates did not realise that a chemical reaction occurs between the sulfur and the carbon in coal, with oxygen from air. Many left them as carbon and sulfur and simply stated carbon and sulfur are given off. Several candidates scored well on this question with carbon dioxide causing global warming and sulphur dioxide causing acid rain as the most common correct explanation. A common misconception is that a greenhouse gas and the greenhouse effect are environmental problems. The problem is global warming and its effects. References to the mining of coal and the effect on the landscape were rare but, nevertheless, gained credit. Comments such as bad for the environment or cause pollution did not gain credit because they do not explain why burning coal causes environmental problems.

The most common response to part (b) scored only two marks for saying that although wood released carbon dioxide into the atmosphere when burnt, the tree had taken in the same amount while it was growing and so it is carbon neutral. Some went on to score the third mark by mentioning that the trees were renewable or could be replanted or sustainable. Most candidates did not appreciate that this neutrality will only work if new trees are planted to absorb the carbon dioxide emitted, else the forest would be completely cut down eventually. Very few candidates appreciate that coal burning releases carbon that has been locked away for millions of years. A common misconception was that wood does not release any or as much carbon dioxide as coal. Additionally a worrying number asserted that wood did not contain carbon so would not cause any pollution. Some candidates thought that carbon dioxide was locked away in trees meaning that when wood burns no carbon dioxide is produced.

6.

- (a) A number of students knew that this was incomplete combustion but far fewer mentioned insufficient oxygen. Some thought the soot was a product of a reaction involving the limestone.
- (b) This item was poorly answered, with a very small number of students able to access all the marks. Many thought that sulfur itself is released to the atmosphere, rather than producing sulfur dioxide in the engine, which is then released. Of those who did, most got the link to acid rain, but a lot did not say there would be less sulfur dioxide and therefore less acid rain. The last point was rarely scored – many just repeated the word erosion or its variants. Very few stated that acid rain reacts with limestone.
- (c) Not very well answered by most students. Many thought that nitrogen comes from the fuel. Others just stated that nitrogen oxides are a product of combustion. Of those who did realise that oxygen and nitrogen were reacting, some thought that this was happening outside the car rather than in the engine and very few mentioned that a high temperature was needed, some just referring to heat.